Title Slide

Project Name: [Hostel Complaint Management System]

Team Name: [Dr.C.N.Rao]

Problem Statement ID: [PS 11]

Problem Statement Overview :

[Hostel Complaint & Maintenance Tracking System (Student-Warden Portal) :
Build a web-based system to manage hostel-related complaints and maintenance requests in a structured and transparent way. Students should be able to raise complaints (e.g., plumbing, electricity, cleanliness, internet, room issues), track their status, and provide feedback after resolution. Wardens/maintenance teams should be able to view and manage all complaints, prioritize them, assign tasks to staff, and monitor pending or overdue issues. The platform must support a clear workflow for complaint handling (e.g., Submitted → Assigned → In Progress → Resolved/Closed), enable priority-based handling (Low/Medium/High), and provide basic analytics to help hostel administration identify recurring issues, resolution time patterns, and backlog/overdue trends.]

Solution Overview:

[A comprehensive, web-based Hostel Complaint & Maintenance Tracking System featuring role-based dashboards, real-time messaging, transparent complaint tracking, and detailed analytics to streamline hostel maintenance operations and improve student satisfaction.]

Solution Details

Proposed Solution

• A modern, responsive web application with separate portals for students and wardens
• Structured complaint workflow with real-time status tracking and priority management
• Integrated messaging system for transparent communication between students and staff
• Analytics dashboard for identifying patterns and improving response times
• Minimalist design with intuitive user interface for ease of use

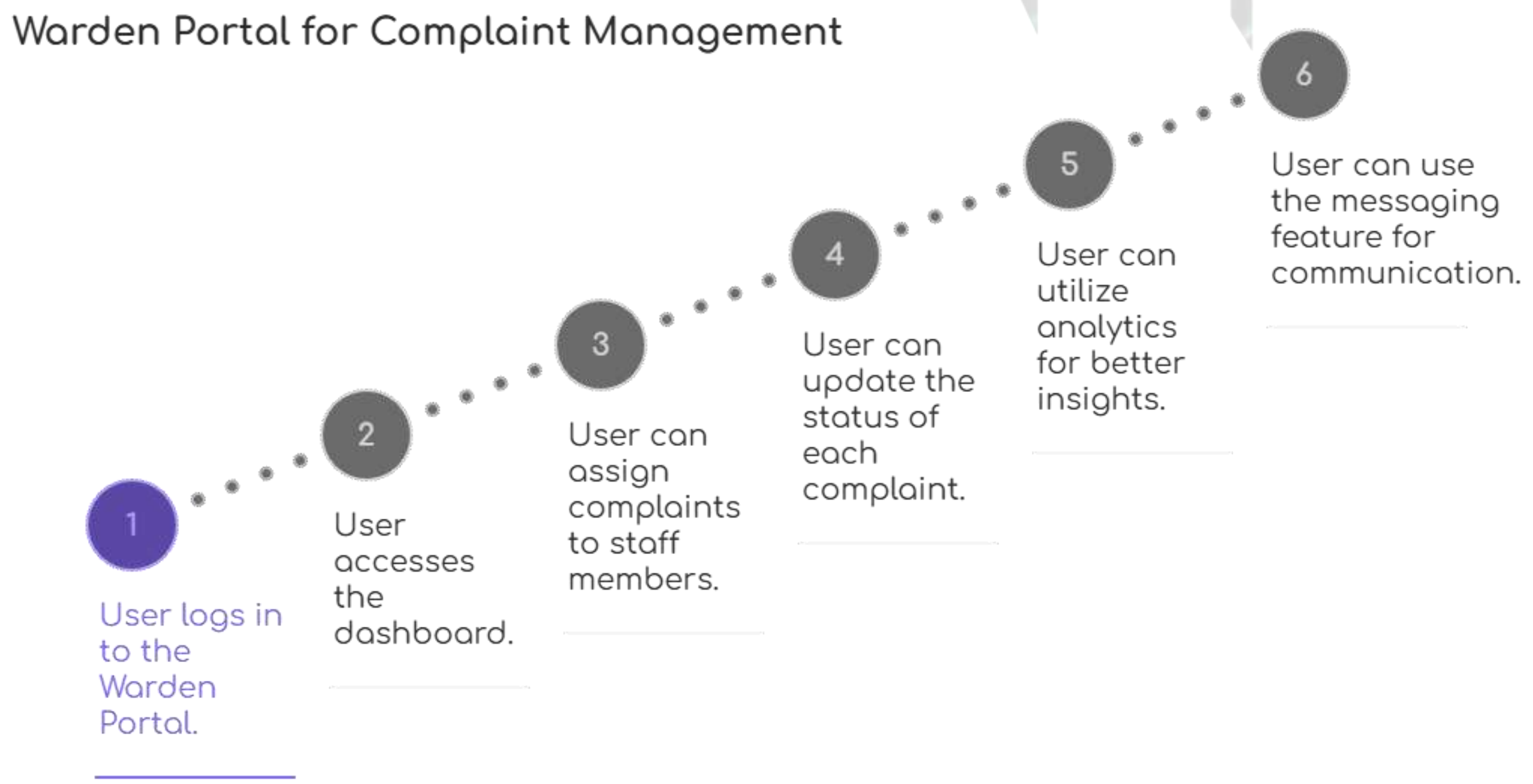
• The system provides students with a platform to submit complaints across multiple categories (plumbing, electricity, cleanliness, internet, room issues) with priority levels and location details. Each complaint follows a structured workflow: Submitted → Assigned → In Progress → Resolved → Closed. Wardens can assign complaints to maintenance staff, update statuses, and monitor progress. The integrated messaging system allows direct communication about specific complaints, ensuring transparency and quick resolution. • How it addresses the problem

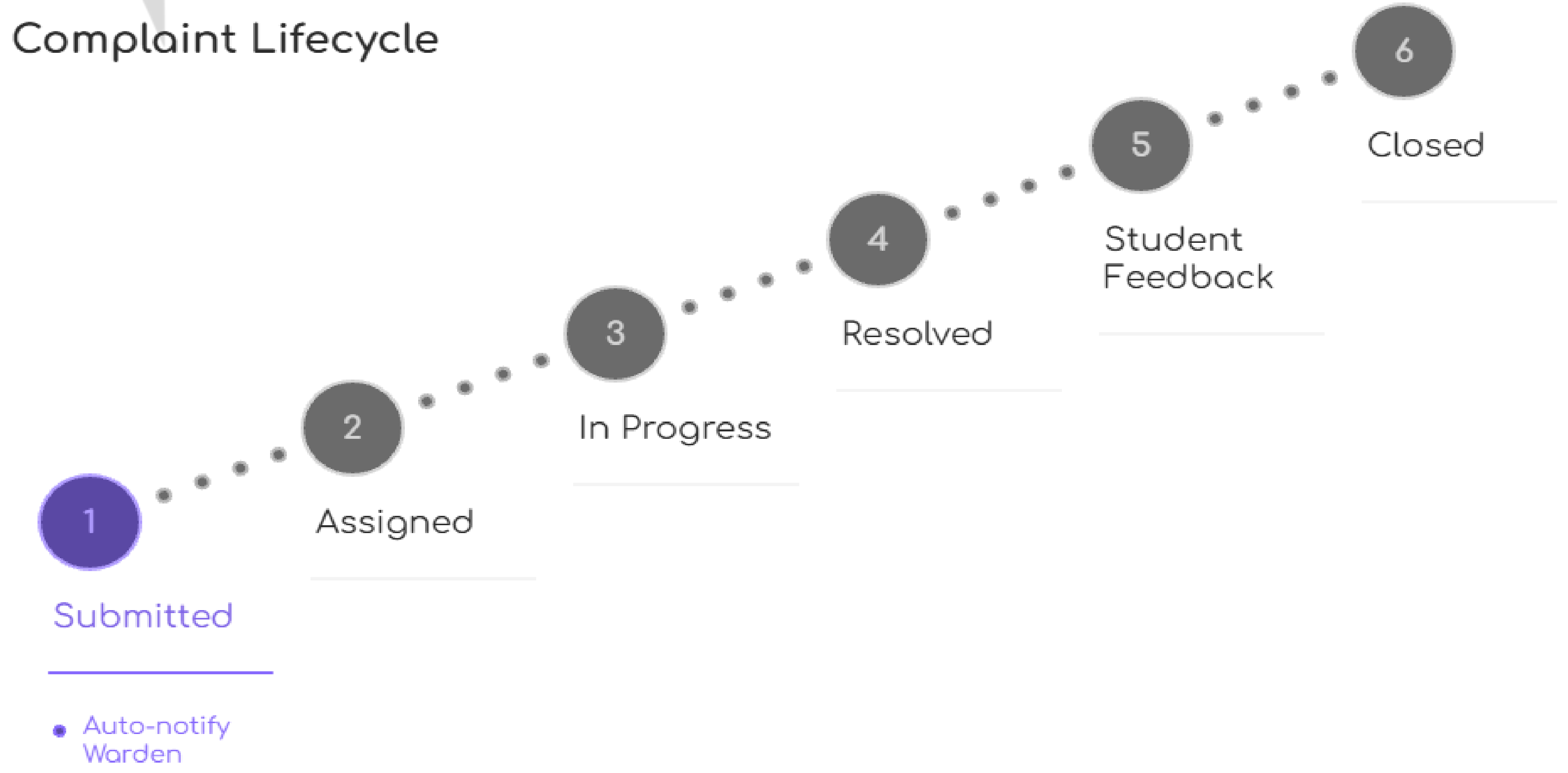
• **Real-time Messaging:** Integrated chat system specific to each complaint
• **Local Storage Database:** No backend server required, easily deployable
• **Comprehensive Analytics:** Visual charts for category distribution, resolution times, and trends
• **Role-Based Views:** Customized interfaces for students, wardens, and maintenance staff
• **Offline Capability:** Works without continuous internet connection

Technical Details

Technologies to be used:
- **Frontend:** HTML5, CSS3, JavaScript (Vanilla)
- **Charts & Visualization:** Chart.js for analytics
- **Icons & UI:** Font Awesome for icons
- **Database:** LocalStorage/IndexedDB for client-side data storage
- **Deployment:** Any static web hosting (GitHub Pages, Netlify, Vercel)
- **Version Control:** Git for collaboration

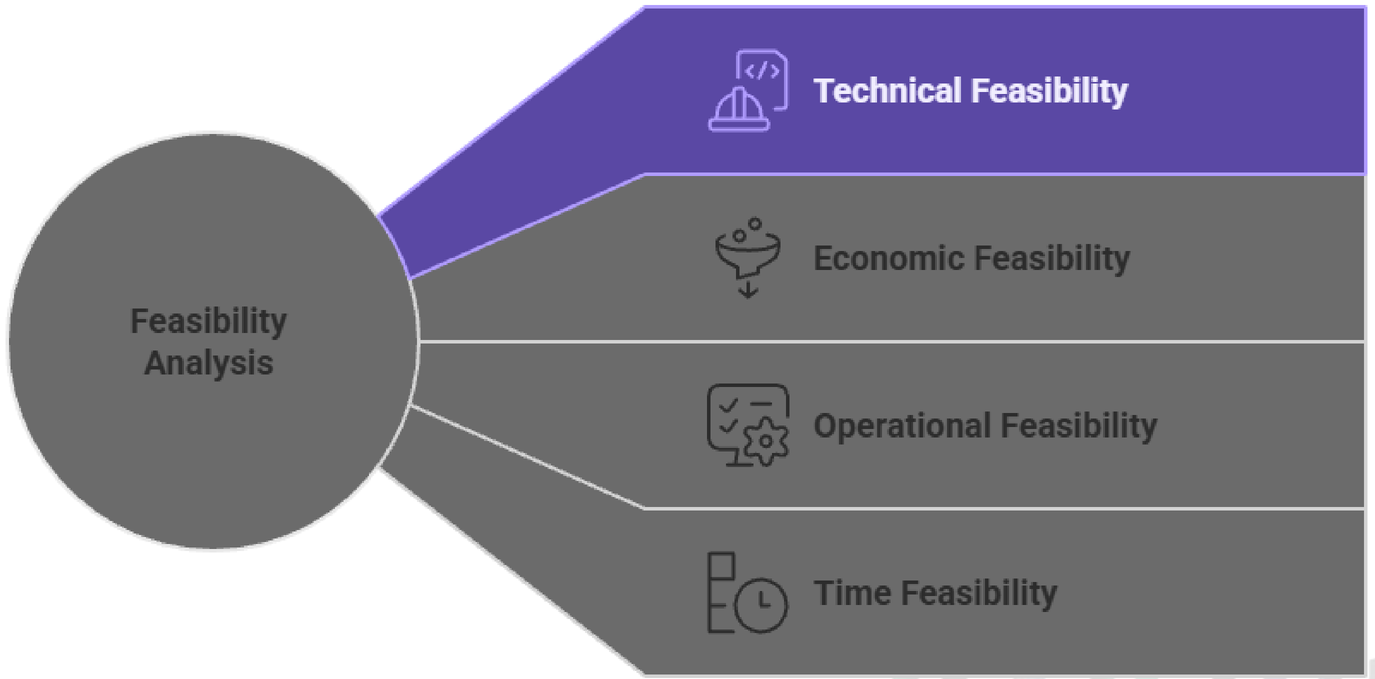
Methodology and process for implementation:
1. **System Architecture:**
2. **User Authentication Layer:** Role-based login (Student/Warden/Staff)
3. **Data Management Layer:** LocalStorage with JSON structure
4. **Application Layer:** Separate HTML pages for each functionality
5. **Presentation Layer:** Responsive CSS with modern UI



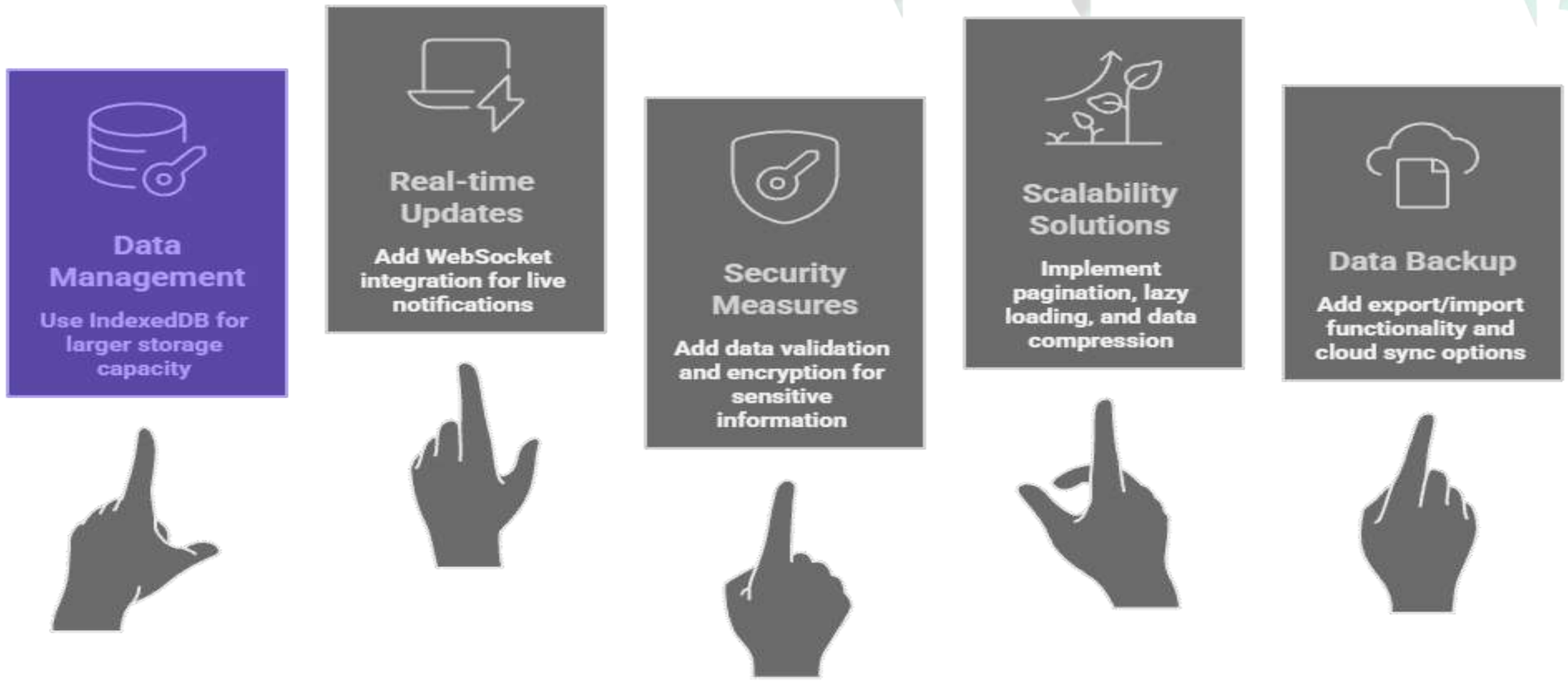


Feasibility & Viability

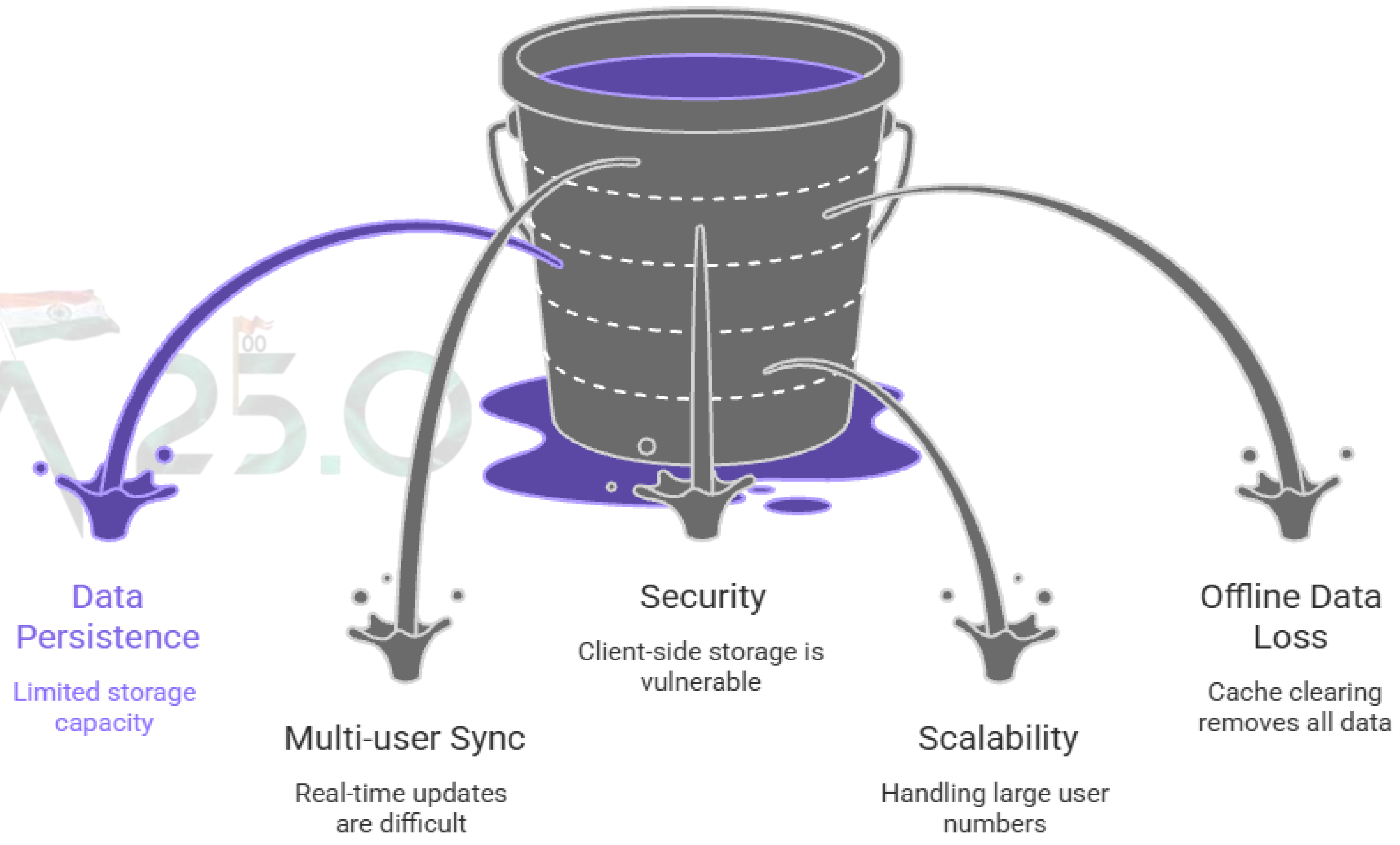
Unveiling the Feasibility of the Idea



How to overcome challenges in the student portal?



Addressing Potential Challenges in Complaint Management



Impacts & Benefits

- **Potential impact on the target audience:**
  - **Students:** Faster complaint resolution, transparent tracking, better communication
  - **Wardens:** Efficient task management, performance monitoring, reduced paperwork
  - **Maintenance Staff:** Clear assignments, priority guidance, accountability tracking
  - **Hostel Administration:** Data-driven decisions, resource optimization, improved satisfaction
- **Social Benefits:**
  - Improved student satisfaction and hostel living experience
  - Transparent communication reduces conflicts and misunderstandings
  - Efficient resource allocation benefits entire hostel community
- **Economic Benefits:**
  - Reduces administrative overhead and paperwork costs
  - Minimizes maintenance delays preventing larger repair costs
  - Optimizes staff utilization through workload balancing
- **Operational Benefits:**
  - Digital tracking eliminates lost complaint records
  - Analytics identify recurring issues for preventive maintenance
  - Standardized workflow ensures consistent service quality
- **Environmental Benefits:**
  - Paperless system reduces environmental footprint
  - Efficient maintenance prevents resource wastage
  - Digital communication reduces physical travel needs

Research & References

- Details / Links of the reference and research work :

https://chat.deepseek.com/share/8nov6gp2idrc1egchs

- Design Links :

https://www.figma.com/make/N3xdwPdoQf56K4ZNjmfk06/Hostel-Complaint-Management-System?t=8ietC087AsHb4uy9-1

➢ Research & References:

- **Competitive Analysis:** Studied existing hostel management software and their limitations
- **Best Practices:** Applied Material Design principles and accessibility guidelines (WCAG 2.1)
- **Technical References:**
  - MDN Web Docs for LocalStorage implementation
  - Chart.js documentation for analytics visualization
  - Font Awesome for icon system
- **Industry Standards:** Followed ITIL framework for incident management workflows

➢ Future Enhancements:

1. Mobile application with push notifications
2. AI-powered complaint categorization and routing
3. Predictive maintenance scheduling based on analytics
4. Integration with IoT sensors for automated issue detection
5. Multi-language support for diverse student populations

➢ Deployment Plan:

1. Phase 1: Basic complaint tracking system (Current Implementation)
2. Phase 2: Mobile app with push notifications
3. Phase 3: IoT integration for automated maintenance requests
4. Phase 4: Predictive analytics for preventive maintenance